# MILITARY SPECIFICATION
# BOOSTER, M125A1E2, LOADING AND PACKING

## 1. SCOPE

1.1 This specification covers the loading, assembly and packing for one type of booster, designated as Booster, M125A1E2, for use in high explosive ammunition.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

| | |
|---|---|
| MIL-B-60687(MU) | - Booster, M125A1E2 Assembly Less Booster Pellet and Lead Cup Charge |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |

DRAWINGS

| | |
|---|---|
| IEL-10520743 | - Inspection Equipment List for Booster, M125A1E2 |
| F10520743 | - Booster, M125A1E2 Assembly |
| F7548175 | - Box, Packing Ammunition for Booster Assembly, Details and Packing |
| C7548174 | - Carton Packing |

(Copies of specification, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC-1390

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3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications.

3.2 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specification and the following:

3.2.1 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

3.2.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.- (Not applicable)

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For classification of defects contained herein, the AQL given for each shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.



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4.3.2.1 Booster subassembly F10520744 covering a detail of Drawing F10520743

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| 1. Booster armed or partially armed | Visual |
| MAJOR AQL = .40% | |
| 101. Insecure, improper or incomplete assembly | Manual and Visual |
| 102. Moisture in flash hole | Visual |
| 103. Foreign matter in flash hole | Visual |
| 104. Booster damaged or defective | Visual |
| MINOR | |
| None defined | |

4.3.2.2 Closing Cup assembly, Drawing B8595506, covering a detail of Drawing F10520743

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR AQL = .40% | |
| 101. Explosive above flush | Gage |
| 102. Explosive below flush exceeds max. | Gage |
| MINOR | |
| None defined | |

4.3.2.3 Booster assembly, Drawing F10520743 (prior to assembly of booster pellet and cup)

| | |
|---|---|
| CRITICAL | |
| 1. Explosive material on small external threads of body | Visual |
| 2. Explosive material on internal thread of booster cup | Visual |



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4.3.2.3 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MAJOR AQL = .40% | |
| 101. Closing cup charge above flush | Gage |
| 102. Closing cup assembly missing | Visual |
| 103. Improper or incomplete crimp | Visual |
| 104. Insecure or improper assembly | Manual and Visual |
| 105. Foreign matter | Visual |
| MINOR | |
| None defined | |

4.3.2.4 Booster Assembly F10520743

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR AQL = .40% | |
| 101. Total length exceeds max. | Gage |
| 102. Staking missing or inadequate (if applicable | Visual |
| 103. Booster cup cracked or split | Visual |
| 104. Booster cup pellet missing (an indication that booster pellet might be missing, if booster cup bottoms on shoulder of body assembly) | Visual |
| 105. Booster pellet moves when assembly is shaken | Manual |
| 106. Threads damaged | Visual |
| 107. Thread sealant missing or inadequate (if applicable) | Visual |
| 108. Dust cover missing or loose | Visual |
| 109. Nomenclature missing, incorrect or illegible | Visual |
| 110. Foreign matter | Visual |
| MINOR | |
| None defined | |

4.3.2.5 Box, Set-up, Packing Ammunition for Boosters, Drawing C7548174, covering a detail of Drawing C7548175.

| | |
|---|---|
| CRITICAL | |
| None defined | |



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4.3.2.5 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MAJOR AQL = 1.0% | |
| 101. Box damaged so that contents are exposed or liable to become exposed | Visual |
| 102. Sealing strip torn, badly wrinkled or incomplete | Visual |
| MINOR AQL = 2.5% | |
| 201. Contents move when shaken | Manual |
| 202. Marking missing, misleading, incorrect or illegible | Visual |

4.3.2.6 Box, Packing Ammunition for Boosters, Drawing C7548175

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR AQL = 1.0% | |
| 101. Board broken | Visual |
| 102. Hardware missing, broken or loose | Visual and Manual |
| 103. Contents exposed | Visual |
| 104. F.S.N. and D.O.D. identification misleading, incorrect or illegible | Visual |
| 105. Strapping missing, broken or loose | Visual and Manual |
| MINOR AQL = 2.5% | |
| 201. Hardware improperly applied or engaged | Visual and Manual |
| 202. Contents move when shaken | Manual |
| 203. Strapping improperly applied | Visual and Manual |
| 204. Seal missing, unsealed or improperly positioned | Visual and Manual |
| 205. Marking misleading, incorrect or illegible | Visual |

4.3.2.7 Process control audit.- Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.



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4.3.2.7 (Cont'd)

a. Booster, M125A1E2, subassembly, F10520743

(1) Removal of dust seal

b. Booster, M125A1E2 Assembly, F10520473

(1) Insertion of closing cup assembly

(2) Assembly of booster pellet and cup

(3) Application of sealing compound to cup threads

(4) Staking operation (if applicable)

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.3.3.1 Binder - lubricant content of tetryl.- A minimum of one sample (8 ounces) to a maximum of five samples (40 ounces) shall be required from each lot to determine the content of binder-lubricant of the tetryl in each lot of boosters. Failure of the sample to meet the requirements specified on the applicable drawings shall be cause for rejection of the lot.

4.3.3.2 Specific gravity of booster pellet.- A sample of ten (10) booster pellets shall be selected from each lot of booster pellets to determine the specific gravity. Failure of one or more booster pellets to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.3.3 Inspection equipment.- The examinations and tests shall be made using equipment listed on IEL-10520743. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Binder-lubricant content of tetryl.- A sample of approximately 8 ounces of tetryl shall be selected and determined as follows: An accurately weighted portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter portions of acetone. Each portion shall be maintained in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up. The side of the crucible shall be washed down with each 10 milliliter portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees (°) to 105° Centigrade (C) for 1/2 hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedure shall be repeated with individual 10 milliliter portions of acetone until the loss in wieght on subsequent extractions totals less than 0.005 gram. The weight of the residue in the crucible shall be calculated to the percentage of total binder-lubricant in the sample to determine compliance with the requirements specified on the applicable drawing.

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4.4.2 Specific gravity of booster pellet.- The representative booster pellets shall be cleaned so as to leave no sharp corners and shall then be accurately weighed on a laboratory balance. To determine the displacement, an apparatus (A8648248) consisting of a glass beaker with ground top, a piece of flat glass of sufficient size to completely cover the top of the beaker, and a semihard rubber perforated plate, pierced at each corner with sharp pointed pins so that above 1/2 inch of pin protrudes from each side, shall be used. The beaker shall be filled with clean mercury, the rubber plate inserted, and the mercury leveled off by pressing down the flat glass until it is even with the top of the beaker. The sample booster pellets shall be placed beneath the prongs of the rubber plate and the mercury leveled as before. The excess mercury shall be caught in a clean bowl. The mercury displaced by the pellet shall be calculated as follows to determine compliance with the requirement specified on the applicable drawing:

$$\text{Specific gravity} = \frac{W}{W_1}$$

where = density of mercury at test temperature, in grams/cubic centimeter

$W$ = weight of booster pellets, in grams

$W_1$ = weight of displaced mercury, in grams

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling and marking shall be as specified on drawing C7548175 and C7548174.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1390-A607



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